

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


# Load dataset

dataset = sns.load_dataset('titanic')


# -----

# Basic plots

# -----


# Histogram using modern seaborn function

sns.histplot(dataset['age'], bins=10, kde=True)

plt.title("Age Distribution with KDE")

plt.show()


sns.histplot(dataset['age'], bins=10, kde=False)

plt.title("Age Distribution Histogram")

plt.show()


# Joint plots

sns.jointplot(x='age', y='fare', data=dataset, kind='scatter')

plt.show()


sns.jointplot(x='age', y='fare', data=dataset, kind='hex')

plt.show()


# Rug plot

sns.rugplot(dataset['fare'])

plt.title("Rug Plot for Fare")
```

```
plt.show()
```

```
# -----
```

```
# Bar plots
```

```
# -----
```

```
sns.barplot(x='sex', y='age', data=dataset)
```

```
plt.title("Average Age by Gender")
```

```
plt.show()
```

```
sns.barplot(x='sex', y='age', data=dataset, estimator=np.std)
```

```
plt.title("Standard Deviation of Age by Gender")
```

```
plt.show()
```

```
# Count plot
```

```
sns.countplot(x='sex', data=dataset)
```

```
plt.title("Count of Male and Female Passengers")
```

```
plt.show()
```

```
# -----
```

```
# Box plots (IMPORTANT QUESTION PART)
```

```
# -----
```

```
sns.boxplot(x='sex', y='age', data=dataset)
```

```
plt.title("Age Distribution by Gender")
```

```
plt.show()
```

```
sns.boxplot(x='sex', y='age', data=dataset, hue='survived')
```

```
plt.title("Age Distribution by Gender and Survival")
```

```
plt.show()
```

```
# -----
```

```
# Violin plots
```

```
# -----
```

```
sns.violinplot(x='sex', y='age', data=dataset)
```

```
plt.title("Violin Plot (Age Distribution by Gender)")
```

```
plt.show()
```

```
sns.violinplot(x='sex', y='age', data=dataset, hue='survived')
```

```
plt.title("Violin Plot (Age, Gender, Survival)")
```

```
plt.show()
```

```
# -----
```

```
# Strip plots
```

```
# -----
```

```
sns.stripplot(x='sex', y='age', data=dataset, jitter=False)
```

```
plt.title("Strip Plot (No Jitter)")
```

```
plt.show()
```

```
sns.stripplot(x='sex', y='age', data=dataset, jitter=True)
```

```
plt.title("Strip Plot (With Jitter)")
```

```
plt.show()
```

```
sns.stripplot(x='sex', y='age', data=dataset, jitter=True, hue='survived')
```

```
plt.title("Strip Plot with Survival")
```

```
plt.show()
```

```
# -----
```

```
# Swarm plots
```

```
# -----
```

```
sns.swarmplot(x='sex', y='age', data=dataset)
plt.title("Swarm Plot (Age by Gender)")
plt.show()
```

```
sns.swarmplot(x='sex', y='age', data=dataset, hue='survived')
plt.title("Swarm Plot with Survival")
plt.show()
```

```
# -----
# Correlation heatmap
# -----
```

```
corr = dataset.corr(numeric_only=True)
```

```
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

```
sns.heatmap(corr, cmap='coolwarm', annot=True)
plt.title("Correlation Heatmap (Coolwarm)")
plt.show()
```

```
# -----
# Fare distribution
# -----
```

```
sns.histplot(dataset['fare'], bins=10)
plt.title("Fare Distribution Histogram")
plt.show()
```

```
sns.countplot(x='fare', data=dataset)
plt.title("Fare Count Plot")
plt.show()
```